

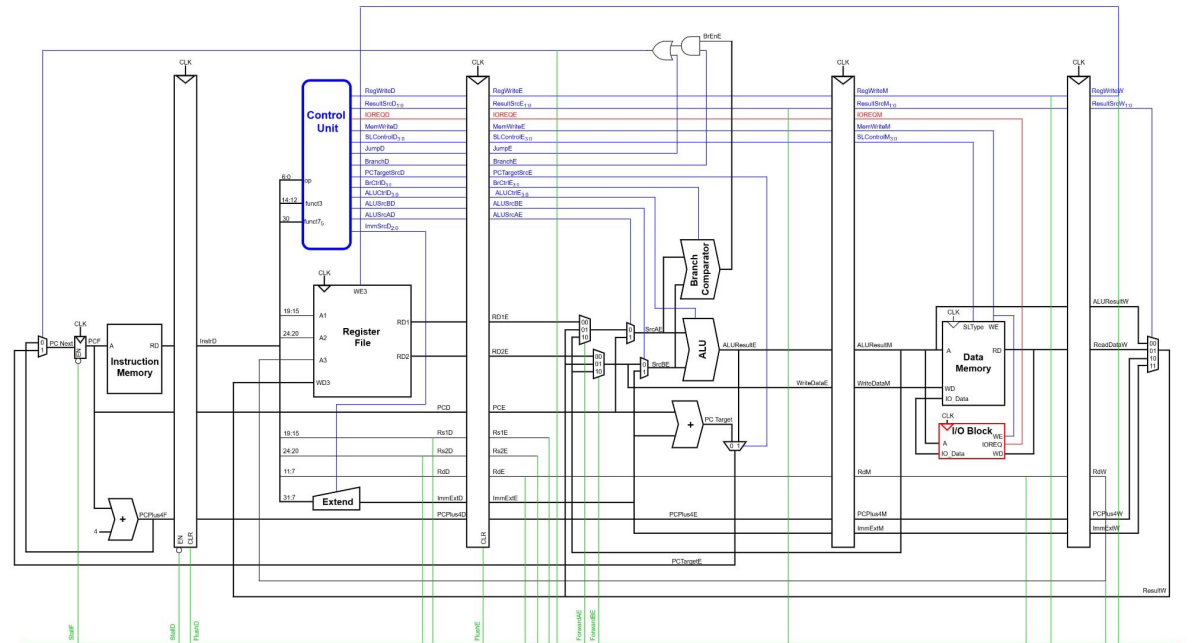
Fetch

Decode

Execute

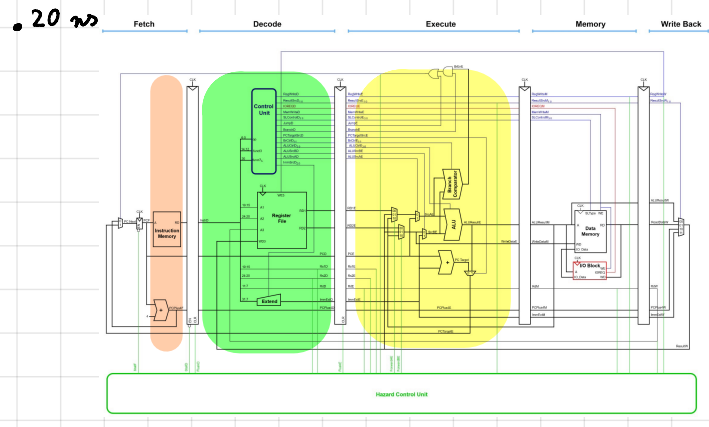
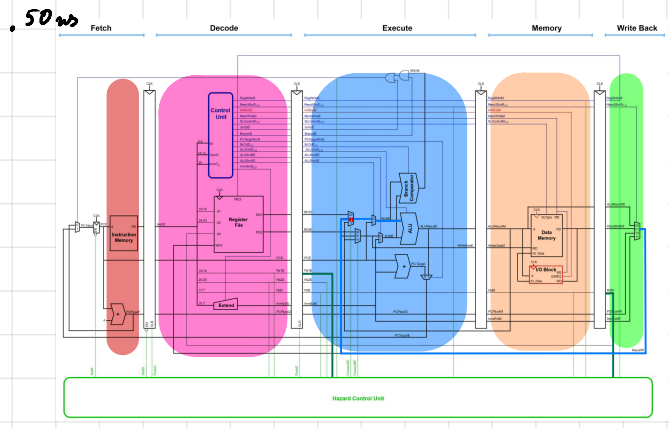
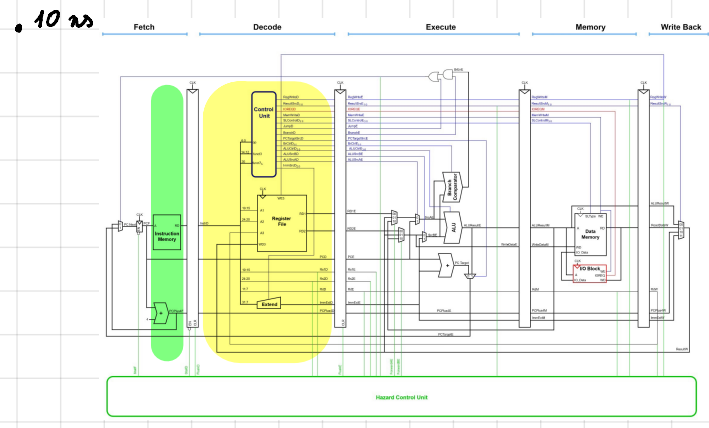
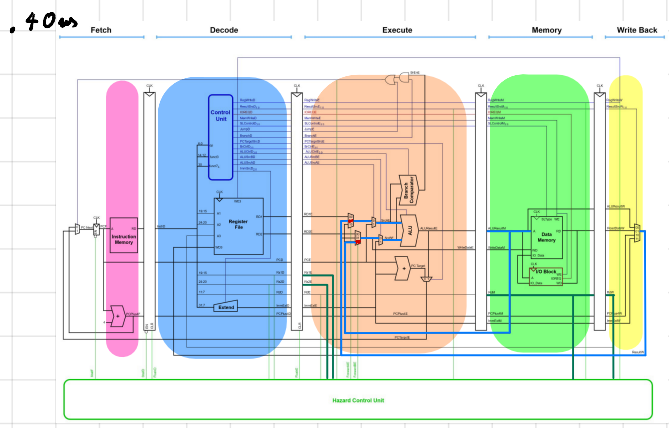
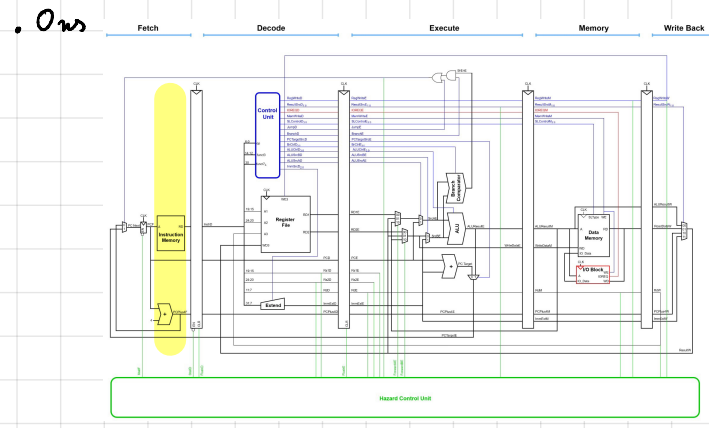
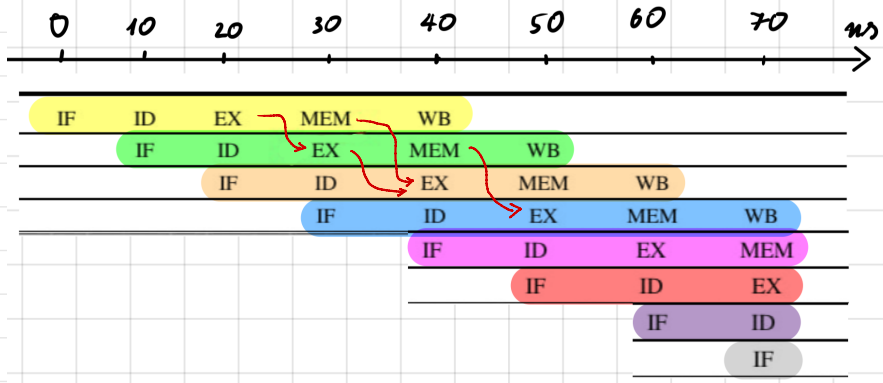
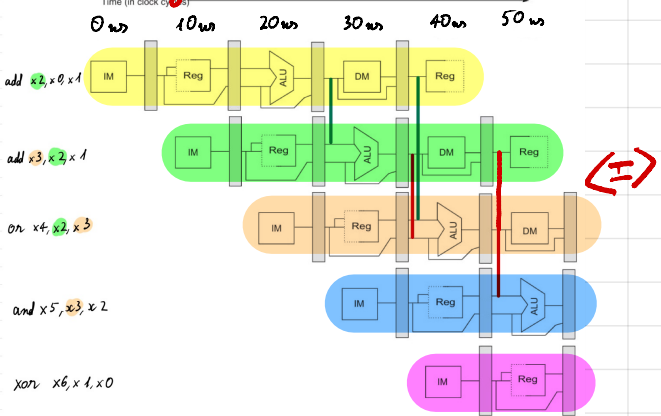
Memory

Write Back



Hazard Control Unit

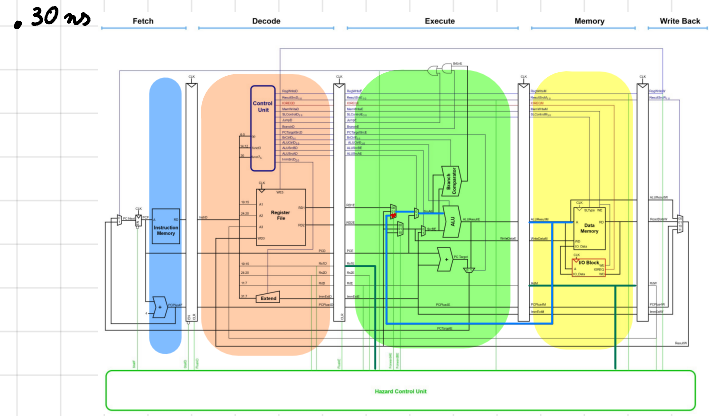
Forwarding



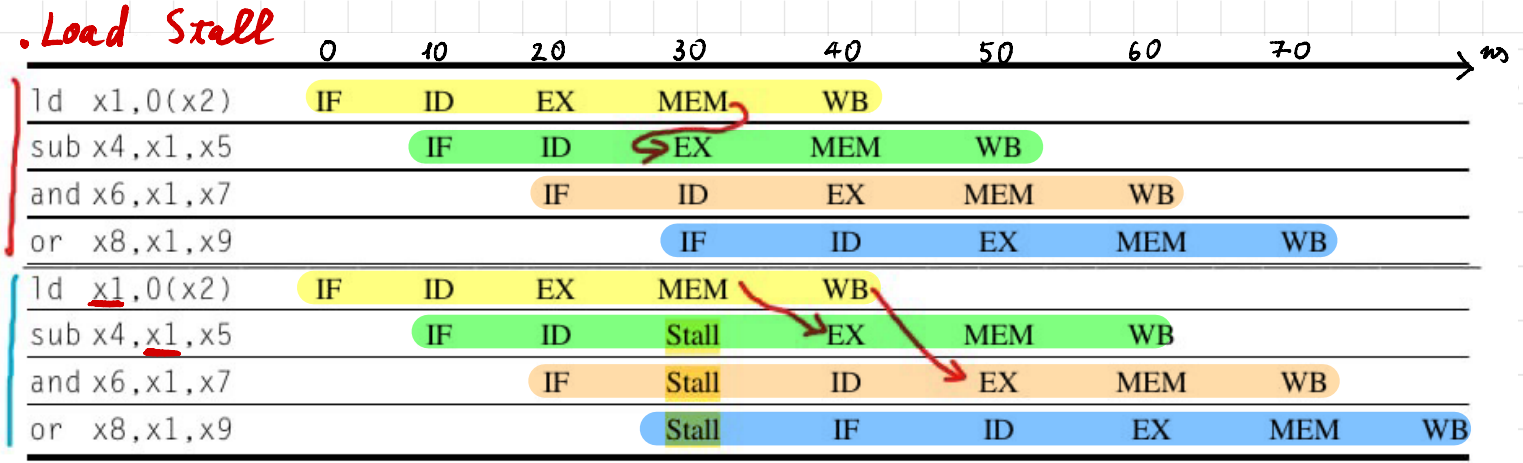
$\Rightarrow$ if $R_{s1E}/R_{s2E} == R_d M$
Forward = 10 ;

$\Rightarrow$ if $R_{s1E}/R_{s2E} == R_d W$
Forward = 01 ;

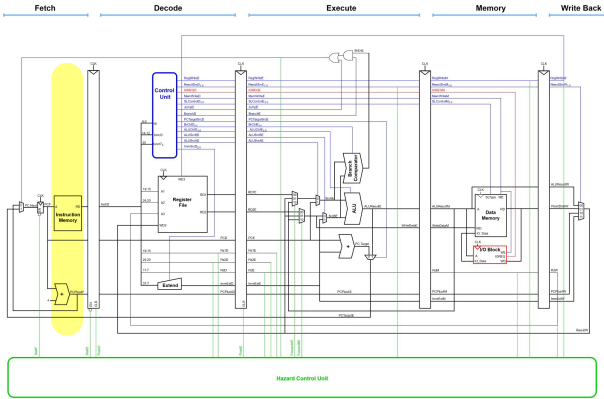
Note: $R_{s1E}/R_{s2E} \neq R_0$
 $\downarrow$
always = 0



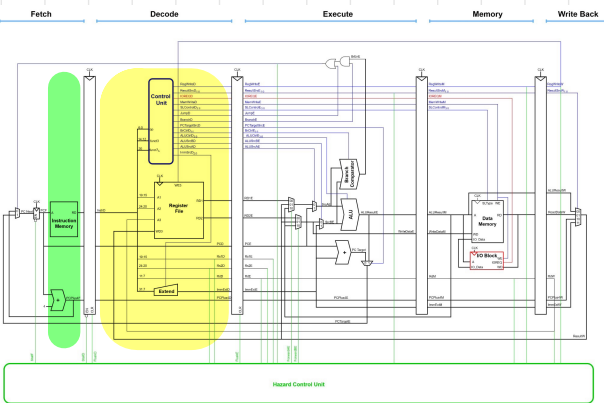
Load Stall



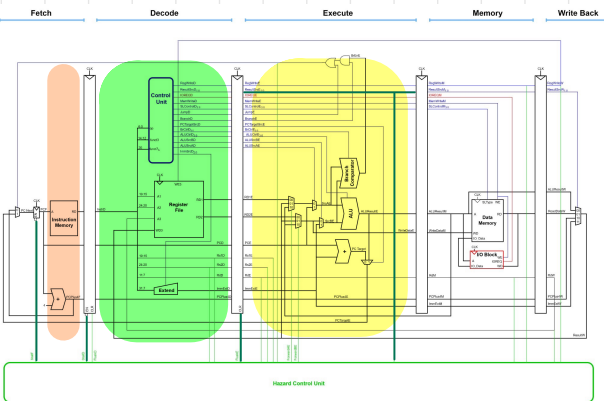
0 ns



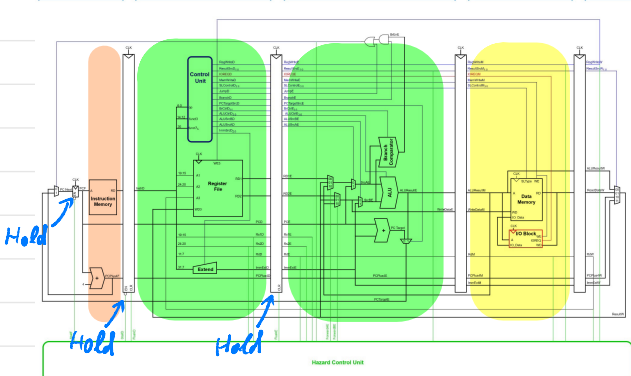
10 ns



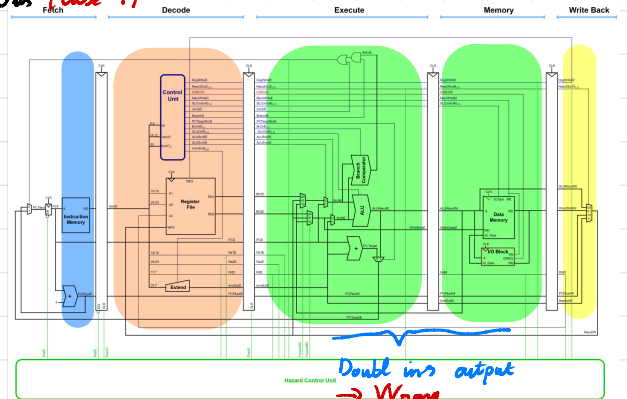
20 ns



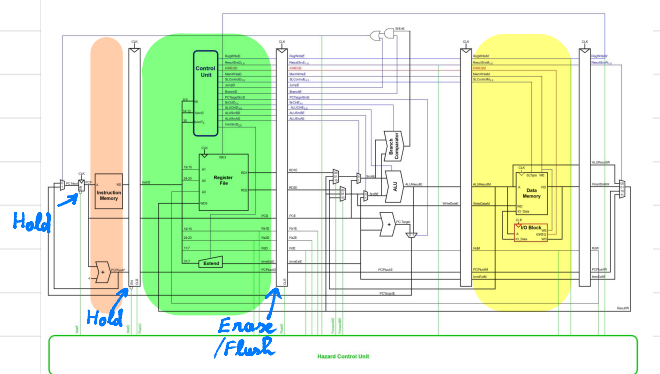
30 ns (case 1)



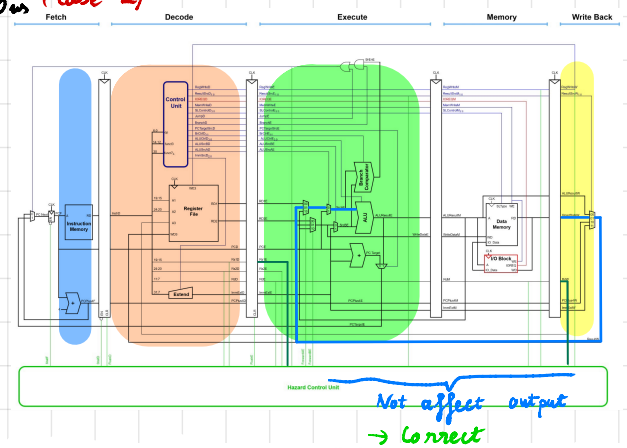
40 ns (case 1)



30 ns (case 2)



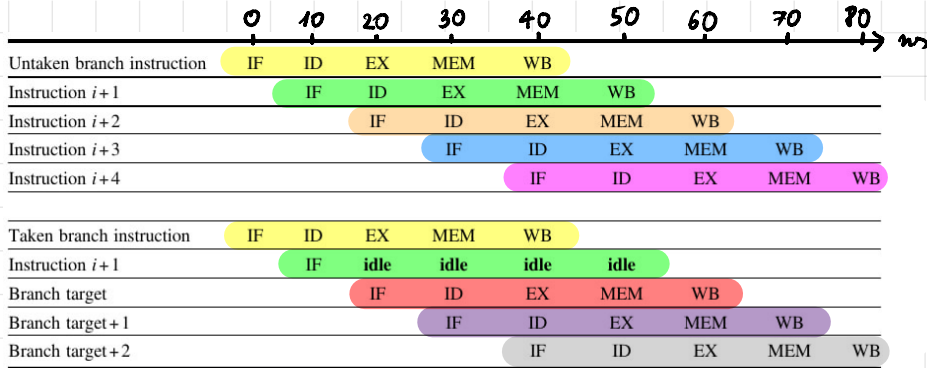
40 ns (case 2)



⇒ if Result SrcE = 01 (Load ins)

⇒ { Stall F = 1
Stall D = 1
Flush E = 1 (at 20 ns)

. At 40 ns: Automatically Forwarding, if RsE match RdW

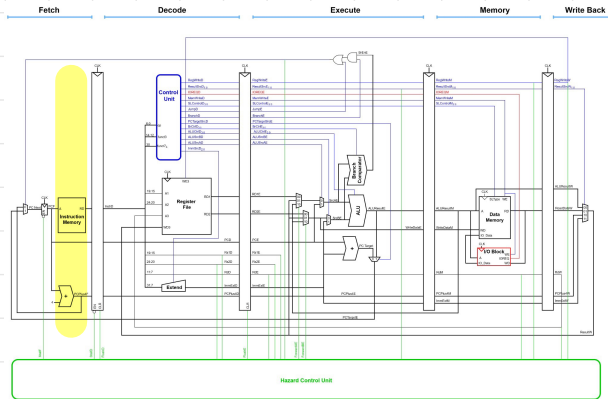


Control Hazard

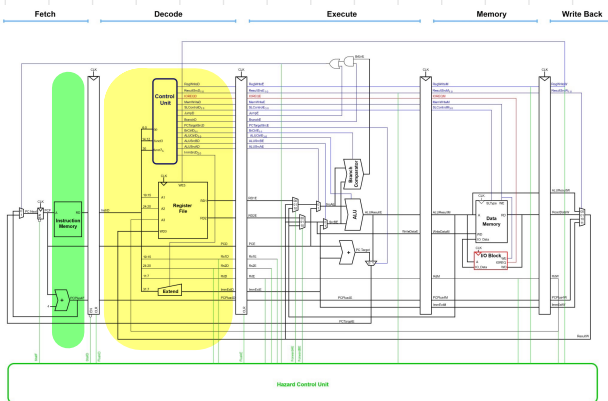
Untaken Case

Figure C.10 The **predicted-not-taken** scheme and the pipeline sequence when the branch is **untaken** (top) and **taken** (bottom). When the branch is **untaken**, determined during ID, we fetch the fall-through and **just continue**. If the branch is **taken** during ID, we **restart the fetch at the branch target**. This causes all instructions following the branch to stall 1 clock cycle.

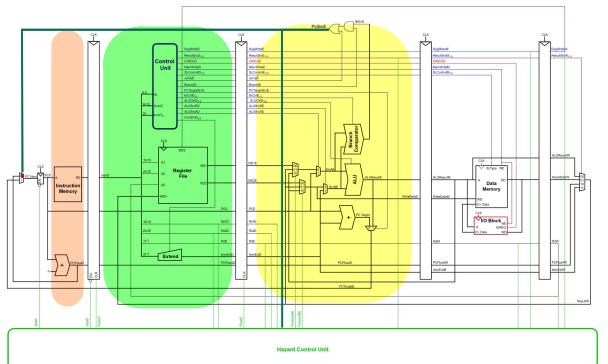
0 ns



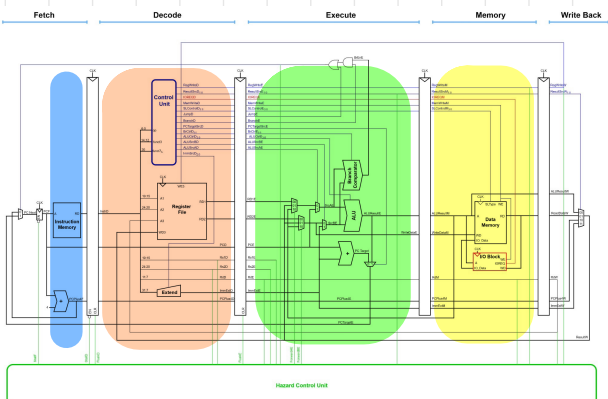
10 ns



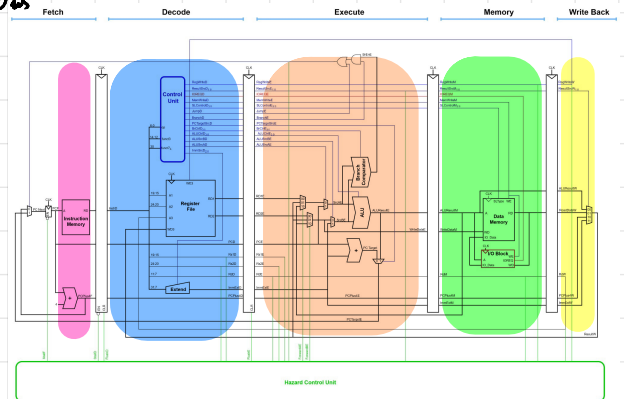
20 ns (branch is untaken)



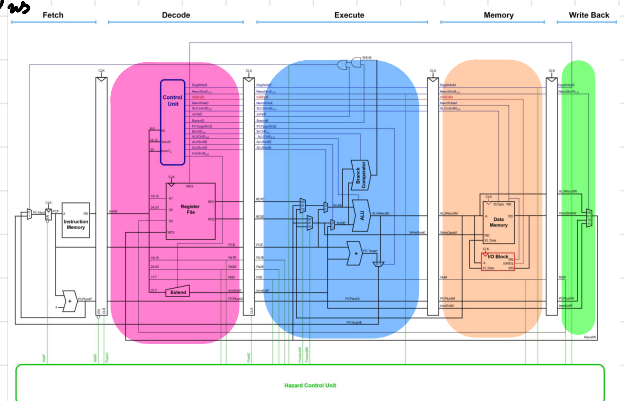
30 ns



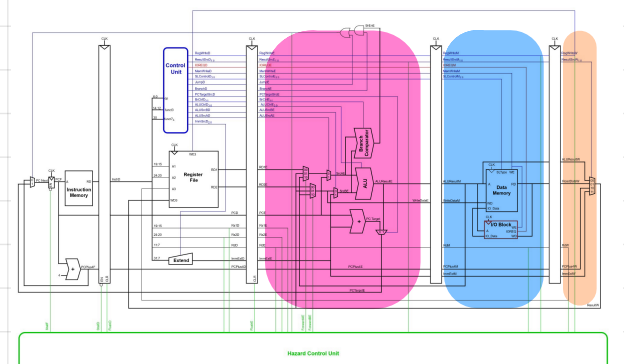
40 ns



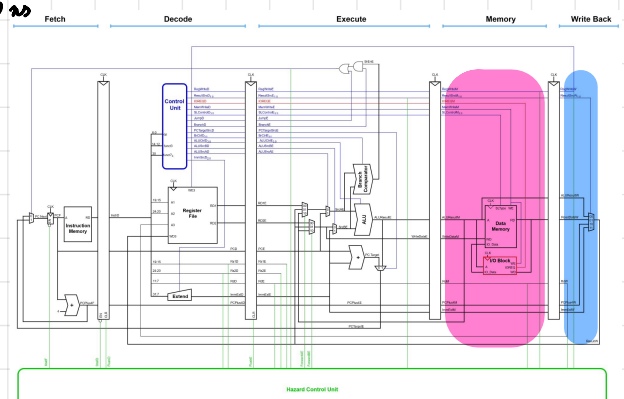
50 ns

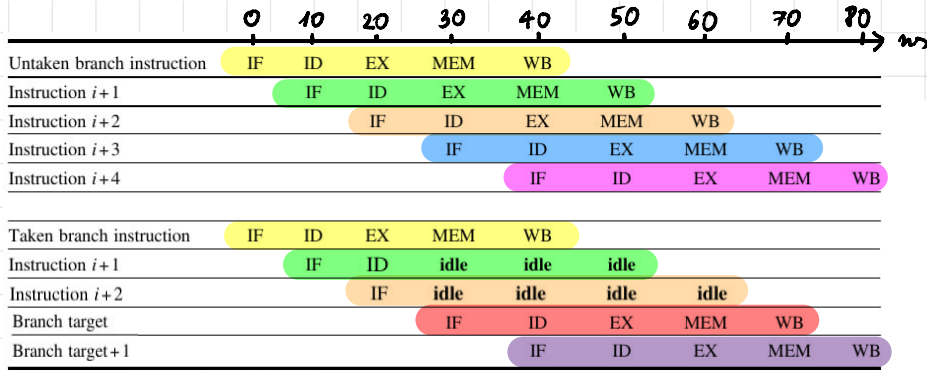


60 ns



70 ns





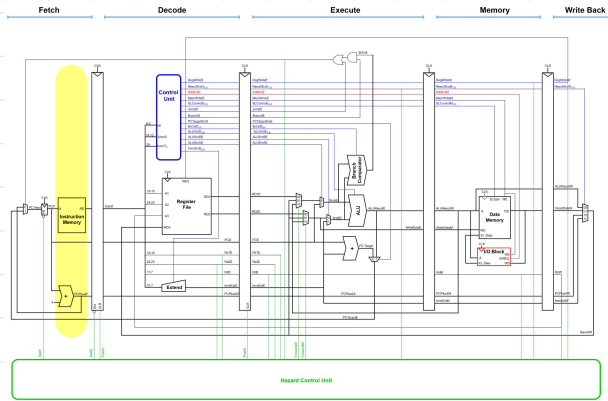
Control Hazard

Taken Case

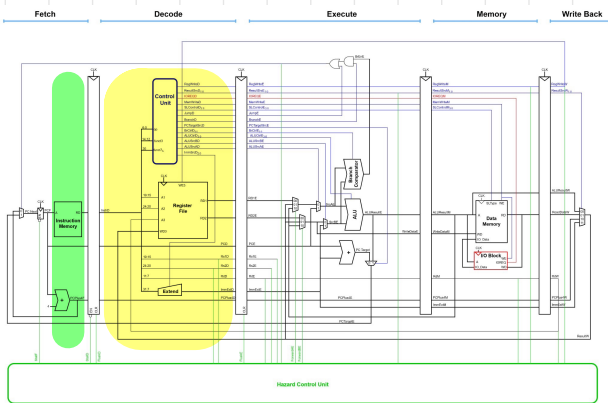
$\Rightarrow$ if $PcSrcE = 1$
 $\left\{ \begin{array}{l} Flush D = 1 \\ Flush E = 1 \end{array} \right.$

Figure C.10 The predicted-not-taken scheme and the pipeline sequence when the branch is untaken (top) and taken (bottom). When the branch is untaken, determined during ID, we fetch the fall-through and just continue. If the branch is taken during ID, we restart the fetch at the branch target. This causes all instructions following the branch to stall 2 clock cycle.

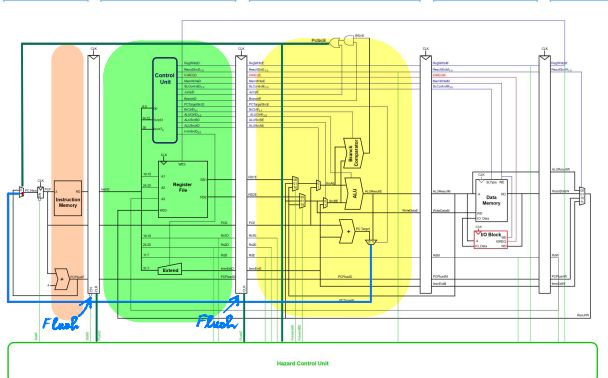
0 ns



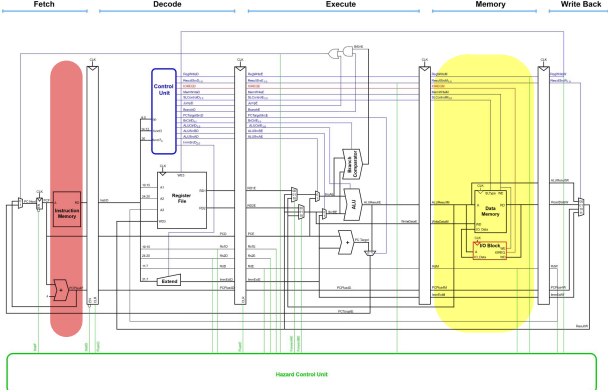
10 ns



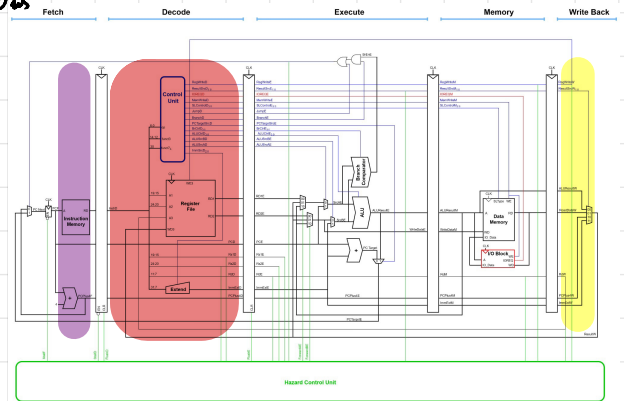
20 ns (Branch is Taken)



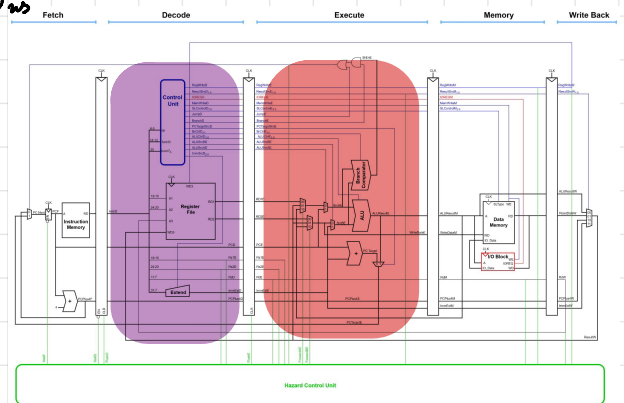
30 ns



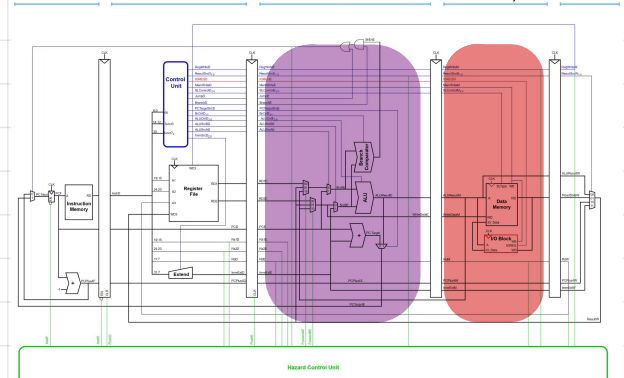
40 ns



50 ns



60 ns



70 ns

